

The empirical recurrence for OEIS A240419, recovered from its tail

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Abstract

OEIS A240419 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 177$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 101$. Its last coefficient is $c_{96} = -136134656 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A240419 is “Number of $n \times 3$ 0..3 arrays with no element equal to two plus the sum of elements to its left or two plus the sum of elements above it or two plus the sum of the elements diagonally to its northwest, modulo 4.”. It has offset 1 and begins

15, 147, 1161, 7857, 52711, 347155, 2331517, 15537761, ...

The entry states:

Empirical recurrence of order 96 (see link above).

As of the “Last modified” line on the live entry (Aug 04 2025 14:54:49, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 177$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 177$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 354$ exact terms from $n = 5$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	14	c_{25}	−420317113685	c_{49}	420210972878766	c_{73}	4611070303421804
c_2	3	c_{26}	−763586300599	c_{50}	41988808876154	c_{74}	1282284974797450
c_3	−881	c_{27}	2507635001731	c_{51}	−392374455936269	c_{75}	−4244556461327889
c_4	3136	c_{28}	−1012015436250	c_{52}	−346467337003459	c_{76}	−171808662786396
c_5	17426	c_{29}	−5533084941919	c_{53}	−325610631481215	c_{77}	3131404068930942
c_6	−114786	c_{30}	7362233310416	c_{54}	1498351840954262	c_{78}	−827512338115490
c_7	−78883	c_{31}	4224513518835	c_{55}	1330041645150455	c_{79}	−1538807395898581
c_8	1828784	c_{32}	−16062699337541	c_{56}	−3353911749483496	c_{80}	908385973219928
c_9	−2029031	c_{33}	9025712954394	c_{57}	−1508648223975934	c_{81}	394033589244946
c_{10}	−15176325	c_{34}	18305746148070	c_{58}	4314066127308262	c_{82}	−461769807445790
c_{11}	37918237	c_{35}	−34975444673057	c_{59}	828394097083677	c_{83}	−1069491694112
c_{12}	54608137	c_{36}	−12858795522277	c_{60}	−2986532617554710	c_{84}	125355648751652
c_{13}	−291995897	c_{37}	69915471898226	c_{61}	−1032607574269317	c_{85}	−24858826611540
c_{14}	130757942	c_{38}	7950666929133	c_{62}	1715357892658681	c_{86}	−16120985180780
c_{15}	1027054728	c_{39}	−105877161858903	c_{63}	1297098325171823	c_{87}	4513882829744
c_{16}	−2267008310	c_{40}	13645505622184	c_{64}	−2906685214018788	c_{88}	−232477237408
c_{17}	619936411	c_{41}	79883382289232	c_{65}	1480793856322139	c_{89}	369699645680
c_{18}	8998105710	c_{42}	−52735304996423	c_{66}	3768308140761700	c_{90}	354031647744
c_{19}	−22508739508	c_{43}	77966844987471	c_{67}	−5494132114655156	c_{91}	−236905088384
c_{20}	−2266036368	c_{44}	6053495595413	c_{68}	−2218959926824530	c_{92}	−46557573504
c_{21}	95513781705	c_{45}	−190394624591090	c_{69}	6490044129174647	c_{93}	20051001856
c_{22}	−118669894704	c_{46}	131348704745461	c_{70}	990052814160725	c_{94}	847432704
c_{23}	−131073743234	c_{47}	−37776520448675	c_{71}	−5293582249208417	c_{95}	−19853312
c_{24}	500040818322	c_{48}	−133360323647595	c_{72}	−1322718926211787	c_{96}	−136134656

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 101$, and it is the recurrence OEIS A240419 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{95} - \dots - c_{96}$. Its constant term is $-c_{96} = 136134656$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -136134656 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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